Fiscal Year:	FY 2026	**Task Last Updated:**	FY 01/21/2026
PI Name:	Chaspari, Theodora Ph.D.		
Project Title:	A Speech-Based Artificial Intelligence System for Predicting Team Functioning Degradation in HERA Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77843-0001	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2024 HERO 80JSC024NA001-FLAGSHIP, OMNIBUS, and NSCOR Crew Health: Topics in Human Health Countermeasures, Behavioral Performance, and Space Radiation and NASA Human Research Program Omnibus Opportunity
Start Date:	01/01/2025	**End Date:**	12/31/2025
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Period of performance changed from 9/25/24 - 9/24/25 to 01/01/2025 - 12/31/2025 per A. Beitman/JSC.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bell, Suzanne Ph.D. (NASA Johnson Space Center)		
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Task Description:

The proposed research aims to build a speech-based artificial intelligence (AI) system that unobtrusively predicts degradation in team functioning, including performance and cohesion, in Campaign 4/Campaign 5 (C4/C5) NASA Human Exploration Research Analog (HERA) missions. The system will conduct automated analysis of prosodic (tone of voice) and linguistic (language content) of speech, model their interaction among team members, and extract influence and turn-taking statistics among speakers, without requiring additional human coding, thus resulting in an automated method that relies on unobtrusive passive monitoring for predicting team outcomes. We will investigate the broad picture of team functioning throughout the whole duration of focal parts of the mission via observing structured and unstructured team interactions. Our research has three primary aims: (1) Pre-process audio recordings from C4/C5 HERA missions, and extract team interaction measures based on acoustic and linguistic features from these missions; (2) Design machine learning models that automatically predict degradation in team functioning using the aforementioned speech measures; and (3) Identify a minimal subset of speech-based measures that are effective predictors of team functioning degradation for each part of the mission, and specific parts of the mission that mostly contribute to team functioning degradation.
Audio data will be pre-processed via noise removal, voice activity detection, speaker identification, and automatic transcription. Acoustic and linguistic features will be extracted, that include: the degree of speech periodicity, shape of the speech spectrum, patterns of vocal intensity, grammatic features, and lexicon-based features capturing cognitive, affective, and social processes. Interaction measures will include the degree of similarity in acoustic and linguistic features among interlocutors, the degree of connectivity of a graph represented by the four team members, and turn-taking measures modeling the conversation progress. Interpretable machine learning (ML) models will be trained based on the interaction measures to predict self-reported team performance and coherence. Ablation studies will be conducted to identify the subset of features, parts of the mission that are most predictive of team functioning degradation, and feasible prediction horizon windows. We hypothesize that a minimal set of speech-based measures, potentially different depending on the structured or unstructured nature of the focal tasks, will be indicative of team functioning degradation, and that the various parts of the mission will contribute differently to this outcome.

Our research will make significant contributions toward the NASA Human Research Roadmap (HRR) Gap "Team-102". It can allow us to isolate and record only the important vocal behaviors during specific parts of the mission, instead of the entire mission, and can be applied to alternate sources of data, such as the ones captured from Artemis, thus enhancing crew privacy. In addition, we examine the predictive ability of speech-based interaction measures without additional sensor devices (e.g., badges measuring crew proximity). Finally, rather than a theory-driven approach, which can be challenging to adapt to space missions, this proposed empirically-driven approach has the potential to identify, in a data-driven manner, behaviors that matter in our specific context.

Ed. Note: For more information on the Artemis campaign, visit: https://

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Results from this research contribute to a greater understanding of factors that affect team operation and performance in isolated and confined environments, and the effective management of future space crews. This work provides new empirical evidence to identify linguistic and acoustic descriptors and parts of the mission affecting astronaut performance, provide unique insights into operationally-relevant team performance, beyond self-report team functioning measures, determine acceptable thresholds to team functioning threats, and inform targeted pre-mission and in-mission intervention strategies adapted to a specific person and/or team, and tailored on evolving needs. The specific research questions and methodologies are developed and tested on HERA data and are specific to isolated and confined environments; thus, beyond space crews, findings and systems developed as part of this research can be extended to future Artemis and Mars exploration missions, as well as Earth teams that operate in isolated and confined environments such as expedition and science teams in the Arctic and Antarctic.

Task Progress:

Aim 1: Pre-process audio recordings from C4/C5 NASA Human Exploration Research Analog (HERA) missions, and extract team interaction measures based on acoustic and linguistic features from these missions
We pre-processed audio data collected from NASA's HERA Campaigns 4 and 5 (C4, C5), involving 9 teams of 4 members each during 45-day analog space missions. Team interactions were recorded across multiple structured and unstructured tasks, including the multi-mission space exploration vehicle-extra vehicular activity (MMSEV-EVA) and the team interaction battery (TIB) task.

We applied an automated audio pre-processing algorithm that combines automatic speech recognition (ASR) capabilities from Open AI’s Whisper software with voice activity detection (VAD) and speaker diarization. Manual correction of the algorithm outputs was performed to ensure the accuracy of the transcripts, including turn-level start and end times and speaker attribution. An evaluation of the automated algorithms for audio pre-processing of the MMSEV-EVA data was conducted to quantify and evaluate auto-generation performance and Whisper transcript accuracy, guide targeted corrections in manual annotation workflows, and assess the feasibility of preliminary speech-based analysis on the auto-generated data. We evaluated three categories of error: Timing Accuracy, Word Accuracy, and Speaker Diarization Accuracy.

Aim 2: Design machine learning models that automatically predict degradation in team functioning using speech measures

We developed machine learning models to predict two primary outcome categories of the TIB. The first category focused on task accuracy in the decision-making task of the TIB. The second outcome category addressed self-reported team functioning, with separate binary classifications for team efficacy and team cohesion. Model inputs included three classes of features: acoustic features, consisting of 118 voice parameters related to frequency, energy/amplitude, and spectral balance extracted using OpenSMILE; lexicon-based linguistic features, comprising 88 descriptors of linguistic, psychological, and cognitive processes derived from the Linguistic Inquiry and Word Count (LIWC) dictionary; and linguistic embeddings, represented by 512-dimensional semantic and syntactic vectors extracted using the Universal Sentence Encoder.

Aim 3: Identifying the most important speech-based measures and mission components

We examined the impact of the different speech-based measures and mission components to the model performance. The results highlight task-specific differences in the feature modalities that are most informative for predicting team

	functioning across the TIB and MMSEV-EVA tasks. We finally evaluated model performance when trained on machine-generated data compared to data that had been further corrected by human annotators.
Bibliography Type:	Description: (Last Updated: 02/06/2026)
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